Li 2019 Melanoma — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Mixed & continuous. At Harmony θ=40 the 3 samples share ONE connected component (0 detached cells), sample- and cell-cycle-mixed throughout. The 2-island split at θ=20 is a technical MARS-seq low-complexity axis (minor island 11/12 markers are MT-pseudogenes / lincRNAs) merged by stronger integration, not deleted.

Gate 2 — module geography → GO (non-branching)

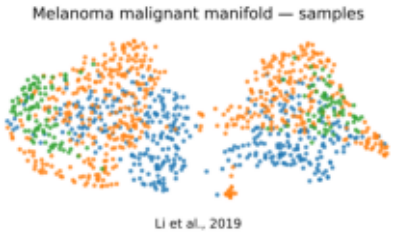
Embryonic pattern present (non-branching). pSom/Gast co-localize at the origin with organized Gast↔pSom anti-correlation (r=−0.61); Crest is an inverse terminal gradient (Crest↔pSom r=−0.59) peaking at the terminus (decile pseudotime 0.80), NOT on a Gast arm — no inversion flag. No bifurcation.

Gate 3 — hub cross-repression → NA

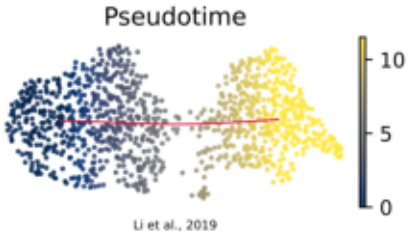
Not applicable — the manifold is non-branching (scFates curve: 2 tips, 0 forks, fold self-approach 2.98; a PPT tree adds no stable fork). With a single axis and no bifurcation there is no hub to test for mutual Gast↔pSom cross-repression, so Gate 3 is NA (branching-only).

STEP FIGURES

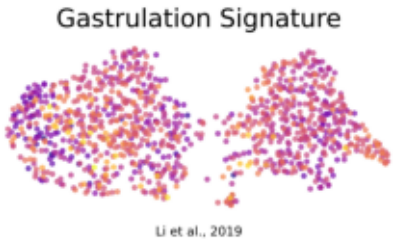
Manifold — samples



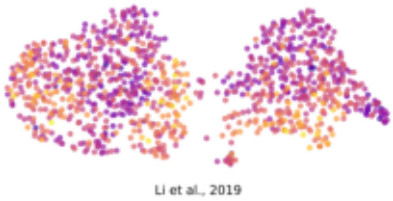
Pseudotime



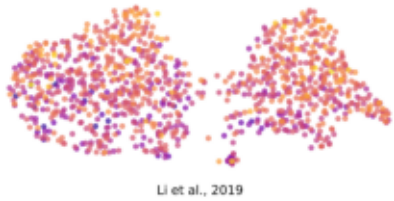
Gastrulation



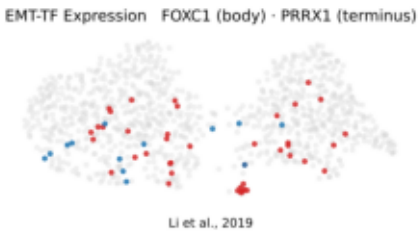
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma (Li et al., 2019), MARS-seq — 1,082 malignant cells (3 samples ≥ threshold); low-pSom, Crest-terminal melanoma (neural-crest lineage).

Verdict.
The embryonic program REPRODUCES as a non-branching axis — a single gastrulation/somitogenesis → neural-crest developmental gradient (Gate 1 GO / Gate 2 GO / Gate 3 NA), the expected melanoma (neural-crest lineage) pattern. EMT-TFs FOXC1 (body/origin) → PRRX1 (terminus).

Parameter choices

Min-cell threshold	1160→1082 cells; dominant cliff 154→21 (ratio 7.3); geom-mean 56 keeps p25/p21/p26, drops 17 spillover	**Integration**	Harmony θ=40 on sample (first strength giving 1 continuous component; θ=20 leaves 2 islands, θ=80 over-fragments); n_pcs=20
Geometry / root	Non-branching scFates curve; root = Gast/pSom-high, Crest-low tip; fold-free (SA 2.98); corr 0.99	**Module means (raw)**	Gast −0.015 · pSom +0.016 · Crest −0.002 (Linear-pSom/Crest)
EMT-TFs	FOXC1 (body/origin) · PRRX1 (terminus); wide panel (Snail/Zeb/Twist + FOXC/PRRX); VIM excluded		